

***Epidemica*: A Modular Digital Platform for Experimental Studies in Human Networks, Behavior, and Digital Twins**

Overview

(Keywords: *experimental epidemiology, proximity sensing, epidemic games, mobile data collection, cohort studies, digital contact tracing, transmission dynamics, digital twins, network science, multi-agent reinforcement learning, external validity*)

In this CAREER proposal, I describe the development and application of *Epidemica*, a modular, open-source research platform designed for high-resolution data collection and experimentation on human behavior, social networks, and pathogen transmission dynamics. To be designed, implemented, and tested by my lab at UMass Chan Medical School, *Epidemica* will enable researchers to create and deploy their own custom mobile apps for collecting behavioral, biological, environmental, and network data in real-world settings. The platform will be built around a customizable, client-server architecture and a suite of interoperable modules that support a broad range of study designs—from seroprevalence cohorts and participatory surveillance to behavioral experiments and randomized controlled trials. By assembling a strong, transdisciplinary group of core collaborators, many of them NSF-funded, we will be able to apply *Epidemica* not only in a number of projects in my own lab, but to empower researchers across the country to conduct field-based studies that generate multi-modal datasets spanning behavior, proxemics, biological testing, and environmental context, and to incorporate detailed early feedback on the strengths and weaknesses of the platform so we can make adjustments to better serve the scientific community.

The proposal is organized around three aims:

Aim 1. Engineer the *Epidemica* platform, including core infrastructure and modular components for proximity sensing, participant onboarding, behavioral interventions, and biological sample integration.

Aim 2. Develop a general framework for validating behavioral and network fidelity of experiments conducted using *Epidemica*, through a multi-site study of voluntary quarantine behavior as a use case.

Aim 3. Integrate *Epidemica* with a novel simulation framework combining agent-based, reinforcement learning urban Digital Twins and synthetic evolving pathogens to explore feedback loops between real-world behavior, pathogen traits, and city-scale responses.

By enabling empirical testing of epidemic hypotheses in controlled yet naturalistic settings, *Epidemica* will transform how researchers study transmission, behavior, and interventions in epidemiology.

Intellectual Merit

Epidemica addresses a core methodological limitation in infectious disease research: the lack of infrastructure to support ecologically valid, experimental studies of human behavior and contact networks during outbreaks. By integrating mobile sensing, behavioral experimentation, and proximity-based pathogen simulations into a unified platform, *Epidemica* will bridge gaps between behavioral, biological, and ecological models. The proposal introduces a novel validation framework to evaluate the realism and external validity of game-based behavior and network data, along with a powerful integrated system that combines epidemic games with urban Digital Twins and genetic epidemiological simulations of pathogen adaptation for exploring cross-scale epidemic dynamics. Together, these innovations will generate mechanistically grounded insights into pathogen spread, behavioral adaptation, and intervention impact—advancing both the science of epidemiology and the methods available to study it.

Broader Impacts

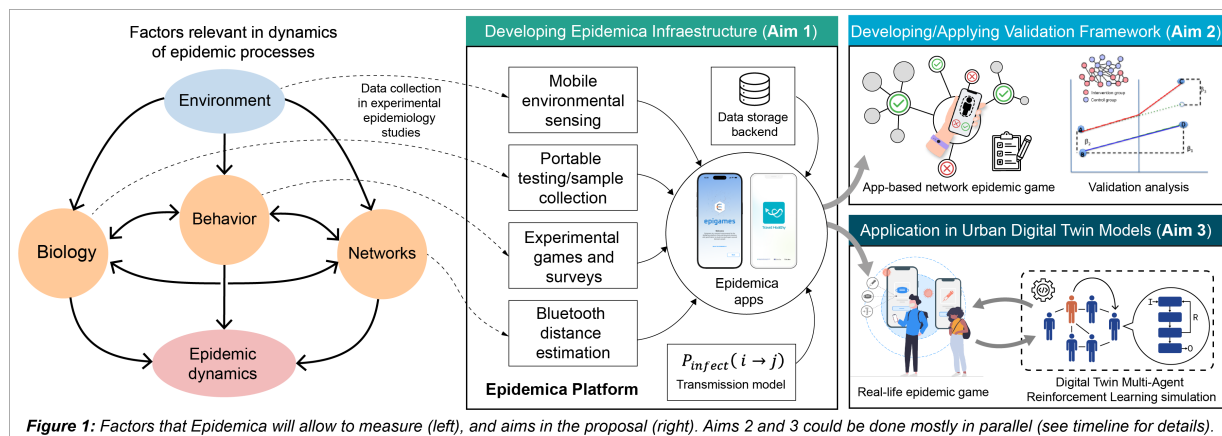
Designed to provide robust and well-documented infrastructure for experimental epidemiology that is available to all Americans, *Epidemica* will democratize participation in infectious disease research. Our user-centered design approach manifested the template *Epidemica* apps, will lower technical barriers, making it accessible to a wide range of investigators, from computational biologists to public health scientists. The platform will support community-based research through partnerships with schools, cultural and social organizations, and international collaborators, and its open-source framework will foster adoption across disciplines. Through training, documentation, and use cases, *Epidemica* will serve as a foundational infrastructure for reproducible science and pandemic preparedness—enabling robust experimental designs, policy-relevant simulations, and public engagement.

1. OVERVIEW

We propose the design and development of ***Epidemica***, a modular digital platform that will serve as shared infrastructure for the research community to collect high-resolution, multi-modal data relevant to infectious disease epidemiology, with a focus on the study of pathogen transmission dynamics and integration with advanced agent-based models (ABMs). *Epidemica* will empower researchers to build custom mobile apps for data collection, enabling the generation of rich datasets that integrate contact networks, human attitudes and behaviors, environmental conditions, and host-pathogen biological information. The platform will support a wide range of study designs—including observational studies, cohort tracking, randomized controlled trials, and experimental games—while incorporating rigorous external validation protocols and realistic pathogen transmission models. This will result in datasets with broad and generalizable utility across biological, epidemiological, and behavioral research domains.

Although digital tools for infectious disease data collection advanced considerably in the past decade—especially during the COVID-19 pandemic [1]—there is currently no comprehensive infrastructure capable of supporting the integrated data collection and experimental capabilities envisioned in *Epidemica*. Our platform addresses this gap by merging interrelated factors that drive transmission, including dynamic contact networks and behavior, which both influence and are influenced by pathogen spread. **Figure 1** (below) illustrates a simplified conceptual diagram linking these factors and the proposed aims.

My lab has strong experience in the development and deployment of mobile tools for infectious disease research. I co-founded the **Operation Outbreak (OO)** project in 2018, which uses proximity-sensing smartphone technology to provide experiential learning opportunities through simulated outbreak scenarios [2]. Our team built the OO platform and led deployments at over 100 schools, engaging more than 10,000 participants. More recently, we developed the **Travel Healthy (TH)** platform in collaboration with the Travelers' Advice and Immunization Center at Massachusetts General Hospital (MGH) [3, 4]. The TH app enables participatory disease surveillance among international travelers. The app's user-centered design was recognized with a 2024 UX Design Award nomination [5], underscoring our lab's emphasis on usability and accessibility. We introduced a first proof-of-concept *Epidemica*-based app, **Epigames (EG)**, at the International Pandemic Sciences Conference held at the University of Oxford in June of 2025 [6].



This CAREER proposal will demonstrate *Epidemica*'s value to the broader scientific community while leveraging its capabilities for a set of foundational projects in my lab. We will carry out, validate, and integrate real-life epidemic experiments conducted with *Epidemica* into a Digital Twins and synthetic biology framework to simulate urban epidemic scenarios incorporating pathogen co-adaptation to human behavior and containment measures. The planned work over the five years of the grant is structured around three specific aims: (1) design *Epidemica*'s architecture and engineer all necessary modules to collect network, behavioral, environmental, and biological data while supporting data quality and FAIR (Findability, Accessibility, Interoperability, and Reusability) principles and working with a core group of scientific collaborators of broad expertise to ensure alignment with the community's needs; (2) develop and apply a rigorous protocol for external validation of behavioral and network data to be collected during *Epidemica*-based experiments; and (3) integrate *Epidemica* with agent-based Digital Twin simulations to model epidemic spread and generate real-time feedback between simulated and real-world data.

2. PROJECT RELEVANCE TO ADVANCING BASIC BIOLOGICAL RESEARCH

Contact networks are a key ecological feature of infectious disease dynamics. Pathogens that spread from person to person, including respiratory viruses such as influenza and SARS-CoV-2, cause a large global health burden and remain the most likely candidates for future pandemics [7]. These pathogens spread through structured, dynamic contact networks [8]—networks shaped by a range of endogenous (e.g., human behavior, host immunity) and exogenous (e.g., weather, housing) factors [9]. Yet, empirical data capturing contact dynamics at high spatial and temporal resolution remain scarce. While disease ecology has explored environmental drivers of transmission, the combined role of biological, behavioral, and environmental contexts—the bio-socio-environmental interface—remains underexplored [10]. Even though properties of real-world contact networks determine not only pathogen spread, but also the effects of interventions and the trajectory of pathogen evolution [11, 12], current epidemiological models rely on idealized network structures (e.g., Erdos-Rényi, scale-free), whose choice significantly alter predictions [13], in terms of key epidemiological indicators such as the effective reproductive number.

There is a persistent gap between the complexity of accurate, policy-relevant transmission models and the granularity of available empirical data. *Epidemica* addresses this gap by offering validated digital infrastructure that enables researchers to measure real-world contact networks, behavioral and spatial dynamics before and after pathogen exposure, simulate gamified disease transmission, and link interaction data with environmental and biological information. Participants may also contribute test results or samples, yielding datasets that connect host-pathogen biology to behavior and environment. These capabilities enable the study of pressing questions in disease ecology and epidemiology, such as: How does contact duration relate to infection risk [14]? Which digital contact tracing strategies are most effective [15]? How do cognitive heuristics emerge during outbreaks [16]? Are there mesoscopic patterns of behavior or mobility that drive transmission [17, 18]?

Epidemica also addresses the absence of a shared experimental infrastructure in infectious disease research. Prior efforts—such as the FluPhone project [19-22], the Copenhagen Networks Study [23, 24], and RFID-based proximity studies [25]—provided foundational data and methods but lacked scalability or long-term support. Contact diary studies like POLYMOD [8, 26] have broad coverage but limited resolution. *Epidemica* builds on this legacy by offering a flexible, extensible platform researchers can reuse, adapt, and share, lowering barriers to replicable and extensible contact-network research.

Human behavior is a critical driver of infectious disease dynamics [27-34], yet few tools exist to empirically evaluate its role in transmission. Experimental games—where participants make decisions affecting their own and others' outcomes [35]—have shown promise in public health, economics, and environmental research [36-41]. Platforms such as oTree [42] have been used in online behavioral experiments simulating epidemics [43-49] but often lack ecological realism. *Epidemica*'s integration of behavioral experiments within naturalistic settings and mobile-enabled data collection offers for the first time a scaffold to systematically test external validity in experimental epidemiology. By combining behavioral surveys, discrete choice experiments, and sensor-driven data collection, researchers will be able to test models of decision-making and behavior under realistic conditions.

Epidemica is also uniquely positioned to support the development and validation of Digital Twin models of infectious disease dynamics. Digital Twins—virtual replicas of real-world systems that update continuously using empirical data—offer transformative potential for biological research [50-52]. By integrating real-time behavioral and contact network data from experimental games into agent-based simulations driven by online reinforcement learning [53], researchers can iteratively refine model structure and parameters based on actual observed responses. This feedback loop not only enhances model realism but also supports adaptive forecasting and intervention testing. *Epidemica*'s ability to bridge empirical experimentation and high-fidelity simulation will open new research avenues into how individuals' networks and behaviors scale to population-level outcomes, and how coupled biological and social systems evolve under epidemic dynamics.

In sum, *Epidemica* will help fill critical empirical and methodological gaps in the field of infectious disease biology. Its infrastructure will enable researchers to collect, analyze, and simulate data that were previously impractical to capture, supporting basic and translational research in epidemiology, computational modeling, and synthetic biology. A key aspect is the possibility of introducing an empirical behavioral layer explicitly into data collection, something that no other approach has done so far.

3. DEVELOPMENT PLAN

Design of the *Epidemica* platform. *Epidemica*'s design is primarily centered around the use of mobile devices (smartphones, wearables) to collect data about social interactions, human behavior, host/pathogen biology, and environmental context in real-life settings to study pathogen transmission dynamics. To support a wide range of data collection procedures and experimental studies, we will design *Epidemica* as a series of stand-alone modules focused on each category of data (contact networks, rapid test results, behavioral data, etc.) that can be integrated into a complete *Epidemica*-based app. A particular app may only use some of the *Epidemica* modules, for example, an app that is meant to sample contact networks of participating individuals and recording seropositivity for a series of respiratory viruses among those individuals during a cohort study, would only need the proximity sensing and biosensing module. Thanks to our experience with earlier app-based projects (the OO and TH apps already mentioned, and other apps such as HandLens [54] that prototyped lateral flow imaging functionality), we will build a template *Epidemica* app to use in our own research studies, and to provide it as an example of how to properly develop a full app with the *Epidemica* modules. The platform will also include backend infrastructure for secure data collection, which we will make available as a self-contained module for easy deployment by other research groups. All these modules and the template *Epidemica* app will be available through a suitable open-source license, and their specs fully documented. The main advantages of this modular design are easier maintaining of the *Epidemica* platform by my team and facilitating the implementation of custom studies and apps by researchers. This modular design even opens the possibility of other groups to contribute new data collection modules to the platform in the future.

Performance metrics. We will evaluate the *Epidemica* platform and promoting its use by the scientific community using three sets of metrics: software quality of the platform, quality and accessibility scores of the data collected with *Epidemica*-based apps, and user engagement metrics from researchers using *Epidemica*-based apps and tools.

- **Software quality metrics.** We will use several metrics to assess software quality in *Epidemica*. Code quality is a critical factor in projects like *Epidemica* that seek to provide building blocks for others to develop their own applications. Quality issues at the code level increase the cognitive load of developers and could therefore hinder platform adoption. There is a large body of work on software quality measurement [55, 56], but many of the proposed metrics have shown weak or conflicting correlation with developers' perception of quality [57]. Based on recent empirical studies [58] and our own experience developing OO, TH, and EG, we will consider readability and structure (internal cohesion, modularity, complexity) as the main indicators of code quality in *Epidemica*. Regular code reviews by our team will use these metrics to inform decisions regarding updates and refactoring of the platform. In addition to those, we will track bug rates, unit test coverage, code security, documentation quality and compliance with suitable code standards [59].
- **Data quality and FAIRness metrics.** For a platform like *Epidemica* to succeed, it must generate high-quality data that can be easily accessed and reused. We are considering accuracy, completeness, and external validity as the initial main data quality measures that we will track over the lifetime of the project [60]. Complete, accurate, and externally valid experimental data is the backbone of scientific research, and the increased digitalization of real-world data collection calls for specific best practices and tools for data quality assessment [61, 62]. Based on the recent literature as well as own extensive expertise, we will compile a set of verification processes in *Epidemica* to ensure continuous evaluation of these metrics, most likely in the form of well-documented study-specific scripts for tailored quality assurance by us and other researchers using *Epidemica*. Just to provide a concrete specific example, one typical dataset to be produced by an *Epidemica*-based app would be a list of contact distances between individuals, estimated using Bluetooth signals. In addition to provide scripts to generate basic descriptive network statistics of this data, we would also provide scripts to calculate the distribution of the estimated distances. As it is known that inter-personal distances follow a bi-modal distribution [63], the output of these scripts will allow researchers to measure how closely the observed data follows those empirical distributions. Due to space limitations, we will not be able to describe our planned quality assurance procedures in this proposal, but it is worth to emphasize that it will be highly-context dependent task. Some metrics, such as external validity, could present unique challenges, some of which we are already aware of (how to generalize attitudinal and behavioral

results generated within experimental games to non-game settings, which is fully addressed in **Aim 2**), and we plan on engaging with our collaborators (see research team and provided LOCs) to find reasonable solutions to these challenges. Furthermore, we will incorporate the FAIR principles for data stewardship [64] so all *Epidemica* modules generate machine-interpretable data and metadata. We will evaluate different protocols, including the resource description framework (RDF), for storing rich metadata and persistent identifiers for all *Epidemica*-generated datasets, and tools and guidelines to automatically evaluate the FAIRness level of the datasets [65], such as FAIRshake [66], FAIR EVA [67], and FAIR Data Maturity Model [68].

- **User experience metrics.** These metrics will be concerned with the assessment of the experiential aspects of use, primarily subjective experience, of the *Epidemica* platform. Users of *Epidemica*-based tools will comprise two distinct groups: researchers applying these tools in their own studies, and participants of those studies. At this stage in the project, our focus in evaluating user experience will be directed towards the first group, researchers (later of stages of development may evaluate participants' experience as well). Thus, we will start by trying to quantitatively and qualitatively measure the experience of researchers in using *Epidemica* in their studies. By applying principles of user experience design and evaluation such as user-centric design [69], as well as theoretical frameworks for users adoption of new technologies, like the Technology acceptance model (TAM) [70], to specific tasks carried out with *Epidemica* (e.g.: building custom study apps using *Epidemica*'s data collection modules, integrating *Epidemica*'s transmission models into simulation pipelines) we should be able to collect several metrics across the following categories: usability (task success rate, time on task), user satisfaction (net promoter and customer satisfaction scores), and user engagement (frequency of use, number of use sessions, session duration). We will gather these metrics through a combination of methods including usability testing, A/B testing, and surveys instruments. We will discuss the evaluation of these metrics during regular meeting with collaborators to assess the need for architectural changes impacting *Epidemica*'s usability.

These performance criteria are directly motivated by our main goal: to provide a foundational toolkit for empirical infectious disease epidemiology, a missing infrastructure in a multidisciplinary field encompassing biology, social sciences, and engineering. As such, the software quality and user experience metrics will be critical to develop and continuously refine the platform, so it does effectively serve the needs of the research community. As to the data quality and FAIRness metrics, ensuring that *Epidemica*-based apps systematically generate data of high accuracy, completeness, external validity, findability, accessibility, interoperability, and reusability would represent a significant contribution to infectious disease research, which has been hindered with shortcomings across all these metrics [71-73]. Ultimately, more realistic models of pathogen transmission dynamics will not be possible until such datasets are openly and reproducibly available to the research community.

Pitfalls and mitigation strategies. While we have confidence in the scientific and technical rationales behind our development plan (the need for an infrastructure platform like *Epidemica*, and feasibility of collecting all data sources listed so far with currently available technology), the following are potential obstacles (and our strategies for circumventing them. First, the initial scope of the *Epidemica* platform, involving four data collection modules (mobile environmental sensing, portable sensing/sample collection, experimental games and surveys, Bluetooth distance estimation) plus a virtual transmission model library and a backend data storage component (**Figure 1**), could be too ambitious given the resources I currently have available in my lab. Second, interest from the research community in adopting *Epidemica* in a meaningful scale could be more limited than our preliminary user surveying has suggested, failing to result in a virtuous feedback loop of adoption and continued development. We propose the following interconnected mitigation strategies for these two pitfalls:

- **Prioritization of development milestones incorporating community feedback.** Complex tool development in a research lab environment should be driven by careful prioritization of milestones and optimized utilization of resources. For *Epidemica*, we will first create a development timeline that considers the tools we have already developed in the lab as initial steppingstones for the larger infrastructure vision. This will allow us to use the gamification approaches, distance estimation algorithm, and transmission model we developed in OO, and the backend infrastructure and data entry components in TH as pieces we can refine to reach, in a relatively short period of time, to four

out of the total six modular elements in *Epidemica* (experimental games and surveys, Bluetooth distance estimation, transmission model, and data storage backend). We will further prioritize these four elements from discussions with our core collaborators, who constitute the first representatives of the larger community. By doing so, we will ensure that the resources in the lab are used most efficiently to satisfy the needs of the community from day one. Refinement and re-prioritization of these initial modules we will be conducted over the lifetime of the project to reflect changing research interests and new insights emerging from our own work with *Epidemica*, or from other relevant projects and initiatives. We anticipate that work on the development of the two new modules of *Epidemica*, mobile environmental sensing and portable sensing/sample collection, modules to start in the second year of the project, but this will also be subject to the continuous assessment and community involvement process.

- Outreach and community-building efforts part of project strategy from first day. My experience in community-driven, open-source technology projects [74-76] makes me acutely aware of the importance of outreach and community-building for the success of such projects. As soon as a minimally usable version of *Epidemica* is available, we will prioritize documenting it extensively, highlighting its first uses by our initial collaborators, and announcing it through multiple channels (official website, conferences, forums, peer-reviewed and pre-print publications, social media, institutional news). These are not minor efforts and need to be conducted systematically, with careful design of message and curation of contents. Based on initial responses, we will evaluate internally and together with our initial core collaborators to refine our outreach strategies. The goal would be to catalyze a first round of community feedback from “second degree” collaborators who would consider using *Epidemica* in their research. This feedback will directly inform the continuous reprioritization of the development work mentioned above.

4. PROJECT MANAGEMENT

The *Epidemica* platform will be designed and implemented by a small and nimble group in my lab. We will follow a version of agile/scrum development [77, 78] we have used for earlier related projects (OO, TH, EG), which proved effective for a team comprising one project manager, developers for frontend and backend work, and user interface & experience (UI/UX) designers:

- Andrés Colubri. Creator of the *Epidemica* platform. Project manager and main scientific and technical lead of the project. He will also direct the outbreak and community-building efforts.
- Andonag Grozdani. Senior mobile software engineer, in charge of architecting and developing the frontend software components of the *Epidemica* platform, including the data collection modules.
- Mansi Khandpekar. UI/UX designer, tasked with conducting user research, creating low- and high-fidelity wireframes and prototypes of the new data collection modules, apps, and adjacent tools
- Backend developers (graduate student + personnel from other funded projects in the lab). The work by these personnel will involve design and implement the new cloud backend, adapting existing pieces from TH and OO, as well as the transmission model library.

This team has worked together for several years on the OO and TH projects, and is adept at using collaborative tools (Trello, Figma, Slack, and Google docs) to enable effective communication. Some development work may be contracted out as other funds become available, including the improvements in the Herald proximity library [79] used for Bluetooth distance estimation (which is already in use by OO). We will leverage automated tools for app testing to perform Q&A at scale [80, 81].

Our methodology will involve monthly sprints to develop new features, followed by internal assessments and discussions with advisors and collaborators, focusing on the performance metrics described earlier. To maximize success of the new backend software engineer position, we will conduct a throughout revision of applications after announcement of the position on UMass Chan job site and other suitable job-posting channels, followed by zoom and in-person interviews. Our team had members joining in the past without any complications. We have developed a collaborative lab culture that is resilient to disruptions (having its origins during the pandemic period) and well adapted to fast-paced projects.

5. RESEARCH TEAM & CORE COLLABORATORS

This proposal is led by PI Colubri, computational scientist at UMass Chan, co-creator of the OO app and creator and lead of the *Epidemica* platform and involves an extended group of core collaborators from

multiple institutions in the U.S. and abroad, with key expertise to advance the *Epidemica* platform. By working with collaborators in various modalities (ongoing joint projects, adoption of the platform, advisory role, analysis of secondary data, etc.) we will be able to demonstrate the flexibility, scientific value, and translational potential of *Epidemica* through a series of targeted use cases that reflect the diversity of real-world experimental epidemiology research. Documenting these specific use cases will provide the scientific community with an initial knowledge base across diverse scenarios, not only showcasing the platform's versatility but also inform the development of guidelines for future users and collaborators. These use cases are designed to provide a representative spectrum of infectious disease research. Each case will validate key technical capabilities of *Epidemica*, while also addressing scientific questions that are important to pandemic preparedness, behavioral epidemiology, and network science. These studies will demonstrate platform versatility across study designs and data modalities and will allow my lab to test and enhance the platform's functionality through iterative development. Crucially, we will be laying groundwork for reproducible and policy-relevant research, by pairing technical infrastructure with rigorously designed, high-impact studies that will show *Epidemica's potential* as a generative platform not only for academic discovery but also for policy-relevant insights.

Each use case will be developed separately with the collaborators who expressed their intent to collaborate in each one of these studies (see provided LOCs). My lab will support their use of *Epidemica* by developing the modules needed in each study to ensure that the data can be generated reliably and reproducibly, with appropriate quality and FAIRness metric assessment. Since the projects have unique specificities within this general framework, below there is a summary for each planned study.

1. Participatory epidemiological surveillance of international travelers (Travel Healthy)

Collaborators: Regina LaRocque (Massachusetts General Hospital)

Objective: To collect illness symptoms and geolocations from U.S. international travelers during their trips abroad to complement traditional public health surveillance methods for travel-related diseases.

Approach: Although the Travel Healthy (TH) app, developed by my lab and launched in November 2023, is already supporting several research studies, the work under this proposal will allow us to formally integrate TH into the *Epidemica* platform. This integration will enhance TH by providing additional development resources and will accelerate *Epidemica* by repurposing TH modules—such as symptom reporting, geolocation tracking, and secure data storage—for broader use. Ongoing work in TH, including support for rapid diagnostic testing for respiratory viruses, will also inform the design of *Epidemica's* biosensor module. This effort will build on our previous work using computer vision to interpret lateral flow assays for hemorrhagic fevers [54], enabling robust self-reporting of symptoms and test results.

2. Seroprevalence cohort study mapping respiratory infections to contact networks

Collaborators: Christophe Fraser, Luca Ferretti, Jasmina Panovska-Griffiths (University of Oxford),

Objective: To prospectively evaluate relationship between contact strength (distance, duration) and risk of respiratory illness infection in small, dense networks (e.g., households, community groups)

Approach: Through field sites established by our Oxford collaborators in Asia and Africa (Singapore, Tanzania), they will design a two-year cohort study to enroll individuals from extended family and local community networks. An *Epidemica*-based app will log contacts and interactions at daily resolution while allowing participants to report serology test results for RSV, flu, and COVID-19. This study is motivated by retrospective analyses from the NHS COVID-19 that showed a linear dependency between contact duration and risk of SARS-CoV-2 infection [14]. The prospective study with follow-up protocols well-defined in advance would provide validation of the earlier results while adopting methodological and analytical safeguards for minimizing confounding and selection biases [82-84].

3. Randomized controlled trial comparing digital contact tracing (DCT) strategies

Collaborators: Christophe Fraser, Luca Ferretti, Jasmina Panovska-Griffiths (University of Oxford), Laurent Hébert-Dufresne, Jean-Gabriel Young (University of Vermont)

Objective: To quantify performance of different DCT methodologies for targeted outbreak control using probabilistic techniques in a gamified outbreak scenario.

Approach: We will collaborate with Oxford and UVM to develop an *Epidemica*-based app using the proximity sensing module to simulate multiple digital contact tracing (DCT) strategies from the literature [15, 85], as previously piloted with the OO app [86]. A synthetic pathogen will spread through participants' contact networks, enabling comparison of three DCT methods: standard tracing (as in the NHS COVID-19

app), simple mean-field (SMF), and belief propagation (BP) algorithms using Bayesian inference. In a randomized controlled trial, participants will be assigned to either a control group (standard DCT) or experimental groups (SMF, BP). Gamified quarantine and testing actions will follow exposure notifications. Outcomes will include infection rates and robustness to variables such as test accuracy.

4. Experimental game probing decision-making and bounded rationality during outbreaks

Collaborators: Tamer Oraby (Texas A&M Kingsville)

Objective: To investigate how bounded rationality and learning strategies shape individual decision-making during epidemics by integrating behavioral game theory and experimental games.

Approach: Dr. Oraby will develop a high-fidelity multiplayer “epidemic videogame” to simulate real-time co-evolution of individual decision-making, peer interactions, and pathogen spread. Built on *Epidemica*’s customizable transmission model, the game will allow experimental control over social and informational variables—such as disease data, network structure, and peer behavior visibility—while capturing detailed behavioral data. Using modeling frameworks that incorporate reinforcement, belief, and social learning, Dr. Oraby will analyze how individuals adapt their strategies based on experience, expectations, and observation. This will enable to quantify heterogeneity in learning and generate empirically grounded insights for improving behavioral models of epidemic response.

5. Experimental game probing decision-making and bounded rationality during outbreaks

Collaborators: Thomas Valente (USC), Laura Moley (Kansas State), Gwen Robbins Schug (UNC Greensboro), Shi Chen (UNC Charlotte)

Objective: To develop and validate a novel modeling framework that integrates experimental epidemic games, diffusion theory applied to adoption of protective behaviors, and network-based SEIR models.

Approach: This is part of a pending NSF IHBEM proposal, where we will combine app-based experimental games with survey instruments and retrospective data to capture behavioral and network dynamics in naturalistic settings. These data will inform an expanded diffusion of innovations model that incorporates multi-level social influence and heterogeneous adoption thresholds, which we will integrate with SEIR transmission dynamics. The framework will be validated through retrospective simulations of historical pandemics and used to generate publicly available synthetic datasets and intervention evaluations. Throughout the project, we will engage public health partners to ensure real-world relevance and deliver a suite of intermediate products, including software tools, policy briefs, and training materials.

5. Observational network epidemiology study examining meso-scale transmission dynamics

Collaborators: John Drake (UGA), Mark Lurie (Brown University)

Objective: To Investigate disease transmission in human systems of varying scales to inform meso-scale transmission theory and generate datasets on mobility and network epidemiology.

Approach: Collaborators Drake and Lurie plan to use *Epidemica* to develop an app to measure how organizational structures affect contact patterns, leveraging case studies (e.g., sports franchises, hospitals, manufacturers). *Epidemica*’s proximity sensing module will allow collection of high-resolution mobility data to discern patterns in human behavior during outbreaks. They will measure how organizational structure influences daily contact, based on case studies facilitated by our stakeholder network, e.g., a major league sports franchise, a hospital system, and a manufacturer. They will consider incorporating synthetic pathogen transmission through these organizational networks, using our gamification techniques developed in other projects. This data will guide the development of a scale-transition theory for transmission where observations at the individual, meso-, and population scales are used in coordination for estimation and inference.

6. Integrating Urban Digital Twins with Synthetic Pathogens in Naturalistic Epidemic Experiments

Collaborators: Soheil Sabri (University of Central Florida), Pablo Cardenas (Cornell University), Arnaud Grignard (French National Research Institute for Sustainable Development)

Objective: To study the dynamic feedback between human behavior, pathogen adaptation, and city-level public health responses.

Approach: The Colubri lab will develop a simulation framework (see **Aim 3** of this proposal) that combines *Epidemica* with genomic epidemiological models and Digital Twin (DT) Agent-Based Models (ABMs) of urban environments to study the dynamic feedback between human behavior, synthetic evolving pathogens, and city-level public health responses, with guidance from Drs. Sabri, Cardenas, and Grignard, experts on urban DT, synthetic biology, and ABMs, respectively. This hybrid platform will enable

real-time, bi-directional coupling between real-world player behavior (via gamified epidemic experiments in naturalistic settings) and a digital replica of the urban system—creating a novel system for exploring pathogen evolution, behavioral adaptation, and systemic responses under complex epidemic scenarios.

6. PRELIMINARY WORK

PI Colubri has an extensive trajectory of building digital tools, from widely used creative environments for interactive art and data visualization [87-89] on desktop and mobile platforms [74] to apps for clinical diagnosis and prognosis [90] using machine learning, but for the purpose of this proposal, the following two projects are the most immediately relevant.

Travel Healthy (TH) app. As part of our broader efforts to build mobile infrastructure for infectious disease research, our lab developed the TH mobile app (**Figure 2**) in partnership with the Travelers' Advice and Immunization Center at Massachusetts General Hospital. Designed to address gaps in traditional travel surveillance—which typically collects data only before and after trips—TH enables U.S. travelers to submit daily symptom surveys and GIS data throughout the duration of their travel. The app also includes traveler tools such as outbreak notifications,

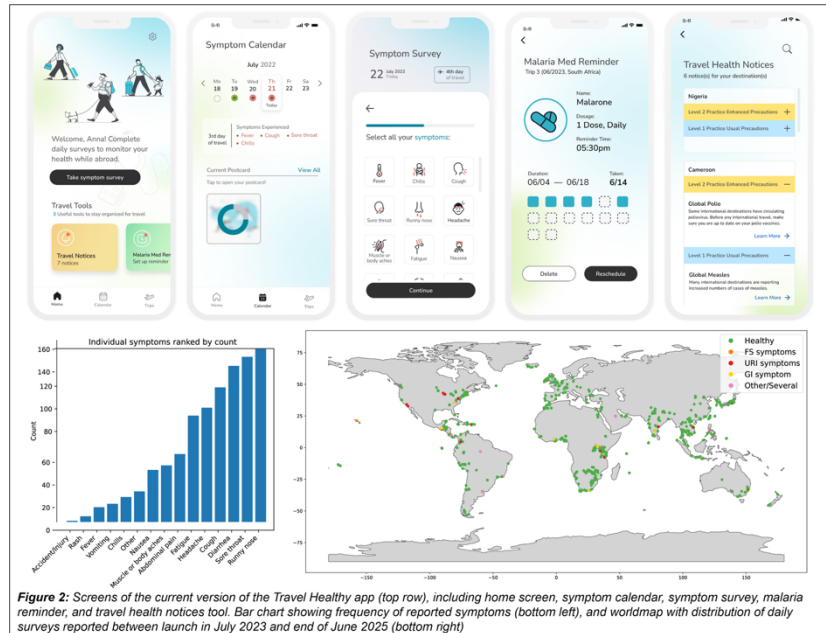


Figure 2: Screens of the current version of the Travel Healthy app (top row), including home screen, symptom calendar, symptom survey, malaria reminder, and travel health notices tool. Bar chart showing frequency of reported symptoms (bottom left), and worldmap with distribution of daily surveys reported between launch in July 2023 and end of June 2025 (bottom right)

a travel document wallet, and malaria medication reminders. Developed using a user-centric design process, TH was enrolled over 300 international travelers since launch in November 2023. Participants submitted more than 5000 daily symptom reports across 800 travel destinations. Nearly 30% of participants reported symptoms during their trip, with a significant proportion being respiratory or gastrointestinal, and GIS mapping revealed geographic clustering, particularly in South Asia. Feedback from the post-travel survey indicated high usability and engagement. This project demonstrates the feasibility and value of participatory, app-based surveillance for real-time monitoring of global health risks and will inform development of *Epidemica* modules for geolocated symptom reporting and travel-linked exposure data.

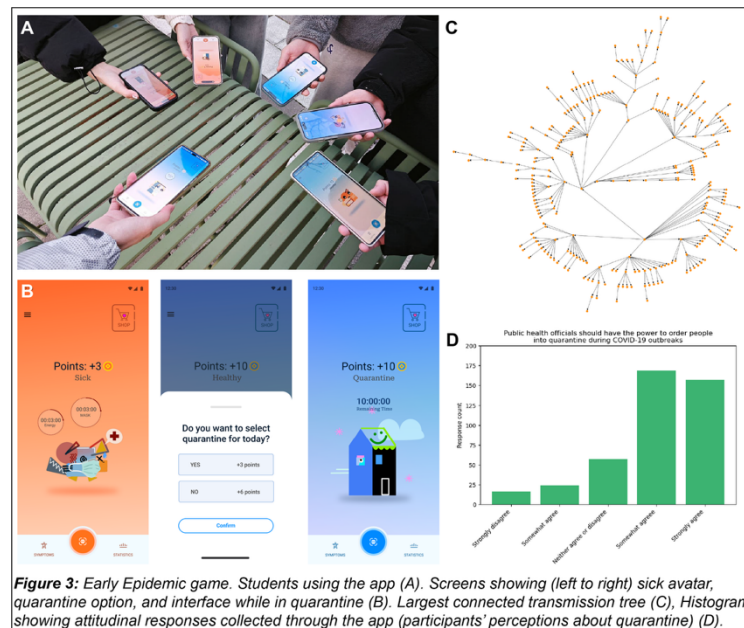


Figure 3: Early Epidemic game. Students using the app (A). Screens showing (left to right) sick avatar, quarantine option, and interface while in quarantine (B). Largest connected transmission tree (C). Histogram showing attitudinal responses collected through the app (participants' perceptions about quarantine) (D).

from the post-travel survey indicated high usability and engagement. This project demonstrates the feasibility and value of participatory, app-based surveillance for real-time monitoring of global health risks and will inform development of *Epidemica* modules for geolocated symptom reporting and travel-linked exposure data.

OO and Epigames apps. Our team has pioneered the development and deployment of smartphone-based epidemic games, initially through the Operation Outbreak (OO) app [4] and now expanded via the *Epidemica* platform and its new Epigames app [43]. These apps enable the collection of high-resolution proximity, behavioral, and decision-making data during simulated epidemic scenarios in real-world

environments. We are conducting a retrospective analysis of 10 research-grade simulations from a larger pool of 72 OO deployments conducted between 2022 and 2023 and encompassing over 7,000 participants. Data from these simulations include detailed contact networks, infection trees, and behavioral metadata, such as participants' use of virtual protective measures within the app. Early analyses reveal that OO simulations replicate key features observed in real-world outbreaks, including right-skewed offspring distributions, superspreading dynamics, and temporal reductions in the effective reproduction number (R_0) over time. Building on OO's foundation, we developed the new Epigames app, designed to scale experiments and enhance data collection. In 2023–2024, a proof-of-concept version of Epigames was piloted across diverse settings, including: Wenzhou-Kean University (WKU, China), ADUSTECH (Nigeria), Murdoch University (Australia), Penn State University (USA), and officially launched at the International Pandemic Sciences Conferences in 2025. The epidemic game at WKU in the Fall 2023, where nearly 1,000 students participated in a two-week epidemic game designed to study the impact of quarantine decisions under conditions simulating trade-offs between individual and collective benefits, is notable because is the largest and longest to date [44] (**Figure 3**). Our analyses demonstrated that even substantial behavioral adherence to containment measures may not yield anticipated reductions in transmission due to complex network effects and behavioral feedback loops. These simulations also reproduced known epidemiological phenomena, such as superspreading, supporting the external validity of the approach.

7. DETAILED RESEARCH PLAN

AIM 1: Development of app infrastructure to collect multi-model infectious disease data

A central objective of this project is to enable researchers to study the transmission dynamics of infectious diseases in real-world settings using mobile sensing technologies. To this end, we will develop *Epidemica* as a modular, scalable, and customizable platform that supports a wide range of research study designs—spanning cohort studies, experimental games, and randomized controlled trials. The platform will facilitate real-time, high-resolution data collection across behavioral, biological, environmental, and spatial dimensions. By architecting the *Epidemica* platform as a well-documented and modular toolkit to create mobile apps to collect multi-modal data relevant for pathogen transmission dynamics, we will arrive to an experimentation infrastructure to advance reproducibility of findings, data quality and FAIR standards, and development of new models in infectious disease epidemiology. Our work in this aim will be informed by our experience of many years with the OO and TH projects, which resulted in apps successfully deployed in multiple settings to collect some of the data we will be generated with *Epidemica*, therefore providing a strong foundation for the project.

Methodology. The development of *Epidemica* will be organized around six core modules and one cross-cutting backend module. Together, they will provide comprehensive infrastructure for conducting complex, field-based epidemiological studies. *Epidemica* will operate under a client-server model: client-side applications (e.g., mobile apps) will be used by participants to collect data using embedded sensors, surveys, and other instruments, while backend services will manage participant registration, study configuration, and secure data storage. Each study-specific app will be built using *Epidemica*'s core libraries and modules, ensuring consistency, ease of development, and interoperability.

Module 1: Study and Participant Registration. This module will comprise a GraphQL Application Programming Interface (API) [91] and cloud-based backend will support dynamic study creation, participant enrollment, and secure authentication workflows. GraphQL is a query language for implementing service-based software architectures, which has been shown to outperform the more traditional RESTful APIs [92]. Using this module, researchers will be able to define eligibility criteria, generate unique study codes, and manage participant groups. Participants in *Epidemica* studies could be optionally linked with external databases for data integration. A web-based management interface will be developed with a suitable JavaScript frontend framework for tracking of participant enrollment and status.

Module 2: Portable Testing and Sample Collection. This mobile-specific module will support collection of biological data using smartphone-based or connected portable diagnostic devices. Its main features will include imaging support for lateral flow assays, Bluetooth connectivity with wearable health monitors, and manual entry of symptoms or test results (adapting current functionality in TH). Drawing on prior work in mobile diagnostics and computer vision [54], we will integrate emerging capabilities for automated

reading and interpretation of rapid diagnostic tests including saliva/blood-based rapid antibody and antigen tests, nucleic acid amplification tests, and dual treponemal/non-treponemal assays [93].

Module 3: Mobile Environmental Sensing. This module will abstract platform-specific APIs (e.g., Android SensorManager, iOS CoreMotion) to enable sampling of environmental parameters such as temperature, air pressure, humidity, and light levels. A unified interface will allow researchers to request sensor data with customizable resolution and sampling frequency. Sensor logs will be automatically timestamped and geo-referenced where appropriate.

Module 4: Experimental Games and Surveys. This module will provide infrastructure for designing and delivering behavioral experiments and attitudinal surveys. Researchers will define parameters for games (e.g., payoff structure, timing, group assignment) and upload pre-defined survey instruments. The current plan is to define a flexible parameter structure to store a variety of behavioral game rules and attitudinal surveys, so they can be stored in the cloud backend for retrieval from the participant's apps. Furthermore, we anticipate that the module will support integration with other modules, such as enabling condition-dependent prompts based on environmental or interaction context, for example, a digital contact tracing game would trigger behavioral choices (e.g.: quarantine or not) based on recent exposure.

Module 5: Bluetooth Proximity and GPS Location Sensing. This module will integrate the open-source Herald library [79], created and developed by one of our collaborators at the University of Oxford, to perform real-time Bluetooth proximity sensing and contact logging. It will also include GPS tracking functionality for capturing individual-level mobility and location data. A unified API will enable simultaneous collection of contact and location data, with real-time data stream preprocessing (e.g., filtering [94], deduplication [95]) handled on-device to preserve efficiency and privacy.

Module 6: General transmission model library. This module will allow simulating realistic pathogen transmission dynamics by incorporating a flexible transmission model to calculate the hazard rate at which the digital pathogen spreads from infectious to susceptible individuals during the epidemic games, allowing to plug-in realistic functions to represent time-dependent infectiousness and individual-level susceptibility, as it is customary in existing ABMs used for real-world disease forecasting [12, 96]. This is a key aspect of our work is to conduct mechanistic realistic simulations of pathogen transmission in naturalistic settings, as part of different *Epidemica* studies. This will be used in all the studies that require spread of a synthetic infectious agent among participants, either using proximity sensing in real-life simulations, in virtual experimentation environments.

Module 7: Data Backend for Researcher-Defined Studies. Researchers will be able to define custom data schemas and deploy study-specific backends on cloud infrastructure (e.g., AWS, GCP, Azure). The backend will manage data validation, secure storage, and access control. APIs will be provided for data export to standard formats (CSV, JSON), as well as connectors for real-time analysis via Jupyter, R, or Matlab. Depending on whether the study requires low-latency and high-concurrency (such as epidemic games with rapid multiplayer interactions) or high-latency and low-concurrency (e.g., periodic analytics, batched reporting, etc.) the backend will switch between a real-time, stateful Phoenix backend server [97] and a burst-load, serverless backend such as AWS Lambda [98].

Outcomes. The modules described above, available as open-source software libraries together with comprehensive documentation of use, will be the main outcome of Aim 1. Another important outcome will be the **Travel Healthy (TH) and Epigames (EG)**, refactored to use *Epidemica* modules and available as template *Epidemica* apps for other researchers, showcasing typical study workflows where participants join a study by entering a unique code, after which the app generates a client-side universally unique identifier (UUID)—a 128-bit value used to track study data without requiring personally identifiable information (PII) for non-human subjects research. Other studies may involve PII based on their own IRB protocols, as it is the case with the studies that use TH and EG. The backend then returns study-specific parameters—for example, transmission model settings for epidemic games—which the app uses to initialize the study on the participant's device. Data is collected via the integrated *Epidemica* modules (e.g., proximity sensing, attitudinal surveys, symptom surveys) and transmitted to the cloud through a GraphQL interface. This interface routes requests to the appropriate backend services for server-side processing and securely stores the data in a PostgreSQL database. Starting with the TH and EG apps as the templates and references for the full *Epidemica* platform will allow us to proceed in a practical way, informed by our accumulated experience and reducing the risk of overengineering (**Figure 4**).

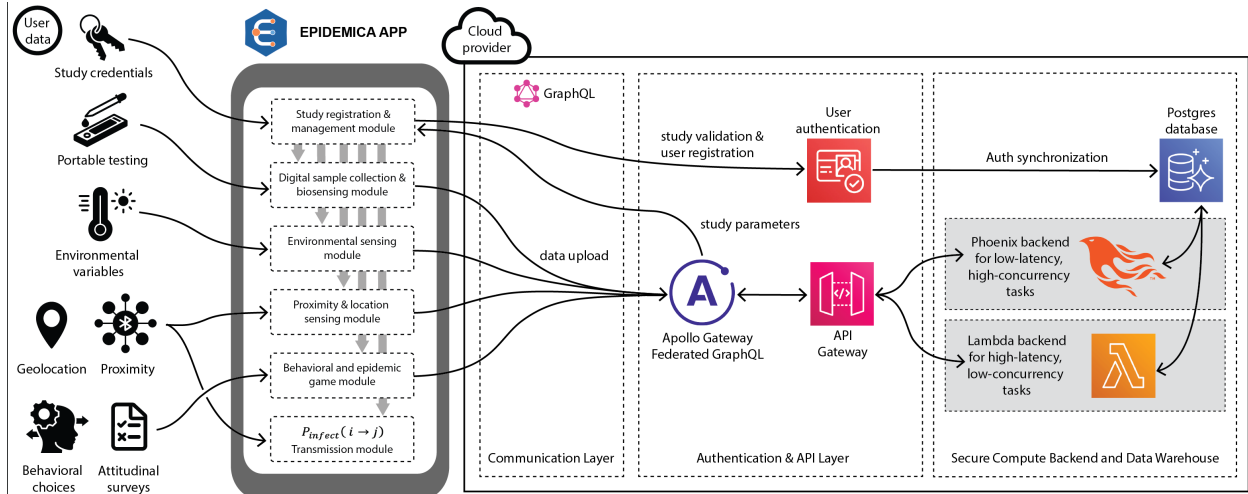


Figure 4: General architecture of an Epidemica-based app, showing all the types of data it can potentially collect through the Epidemica modules available in the app itself, and the cloud backend infrastructure supporting data storage.

AIM 2: Develop and apply a general framework for evaluating the behavioral and network fidelity of Epidemica-based experiments

Understanding the behavioral dynamics and social networks that influence pathogen transmission is critical for developing accurate epidemiological models and effective public health strategies [27-29]. However, most behavioral research in epidemiology relies on observational surveys, self-report instruments, or artificial laboratory settings, limiting the external validity and ecological realism of the resulting data [30]. *Epidemica's* app-based experimental platform offers a novel opportunity to collect behavioral and network data at high spatiotemporal resolution, embedded in realistic, game-based simulations of infectious disease spread. To realize the potential of this experimentation approach, however, there remains the need to validate the data collected through such gamified experiments, as decisions within the game are typically operationalized through simple digital actions, lowering the physical, social, and psychological costs experienced in real-life contexts. More specifically, we must establish to what extent in-game decisions reflect real-world attitudes and behaviors, and whether network dynamics captured by mobile sensing reflect actual patterns of social interaction relevant to disease spread. We address this need by developing a general validation framework for *Epidemica*-based experiments, using a multi-site study focused on voluntary quarantine behavior as a use case.

Methodology. We will implement a structured validation pipeline for behavioral and network data across two components: behavioral (or indicator) parallelism [99], and network fidelity [100]. This framework will be developed and tested through a series of app-based “epidemic games” conducted in multiple institutional and cultural settings (e.g., universities, community settings, participants of health research programs). The central case study will be the **Quarantine Game (QG)**, a scenario-based simulation in which participants face decisions about whether to self-isolate after exposure to a simulated digital pathogen, with varying levels of peer behavior information and in-game incentives (study protocol for QG approved by UMass IRB). Due logistical and ethical reasons, participants do not actually isolate during the QG but chose the quarantine behavior by pressing a button in the app and receive point rewards or penalties depending on their choices and eventual simulated outcomes within the game (“infection”, “death”). The study protocol comprises the following stages: (1) recruitment, during which participants install the *Epidemica* app and complete baseline attitudinal surveys, (2) calibration, when participants complete parallel surveys reframed in the game context, (3) group assignment, via graph-cluster randomization methods [101] to minimize cross-talk between groups, (4) experimental manipulation, where each group is exposed to different experimental conditions (e.g., presence or absence of information about peer behavior; aggregate vs. individual-level peer compliance data, local vs global level of risk or number of cases), and (6) data analysis comprising ANOVA for NxM factorial study design, difference-in-differences, and triangulation analysis. For the external validation part of the pipeline, we will

evaluate behavioral parallelism and network data fidelity, with the goal of arriving to a general index of external validity that can be applied in other experimental setups (**Figure 5**). Dr. Dmitri Williams (the first researcher to use online games as behavioral experimentation tools [102], LOC provided), will assist in the development of this index, in conjunction with Dr. Nick Bowman (expert in social impacts of interactive media, including gaming, LOC also provided) who will help in defining the survey instruments.

1. Behavioral parallelism. This involves the game-to-real-life mapping, which will be quantified by comparing recruitment and calibration stage survey responses (e.g., attitudes toward quarantine, perceived efficacy, self-efficacy, and perceived social norms) and the attitude-behavior gap to assess the alignment between expressed attitudes and revealed behavior, which we will also test for concordance between survey responses and decisions made under different game conditions.

2. Network Data Fidelity. For the underlying proximity sensing, we will use smartphone Bluetooth-based proximity sensing to infer physical interactions among participants during gameplay. To evaluate the accuracy of these interaction networks, we will conduct small-scale observational calibration sessions (e.g., during classroom or orientation events) and compare inferred interactions against contact logs by participants and researcher observations. To quantify network consistency, we will assess the stability of the contact network over time and across settings, using standard network statistics (e.g., degree distribution, clustering coefficient) and comparing them to existing contact network studies (e.g., RFID-based datasets from SocioPatterns project [103], POLYMOD [8, 26]).

External Validity Index: We will synthesize the results of the above analyses into a multi-component **External Validity Index (EVI)** that captures the alignment between game-based and real-world behavior. The EVI could combine indicators of attitude-behavior consistency (survey vs. choice data), behavioral responsiveness to social cues (experimental manipulation effects), network accuracy (compared to observational baselines through a suitable concordance measure such as the Jaccard index), and cross-site generalizability (replication of results across institutions and populations). This index will be made available as an open-source tool and accompanying documentation package, allowing researchers to benchmark the validity of their own *Epidemica*-based experiments.

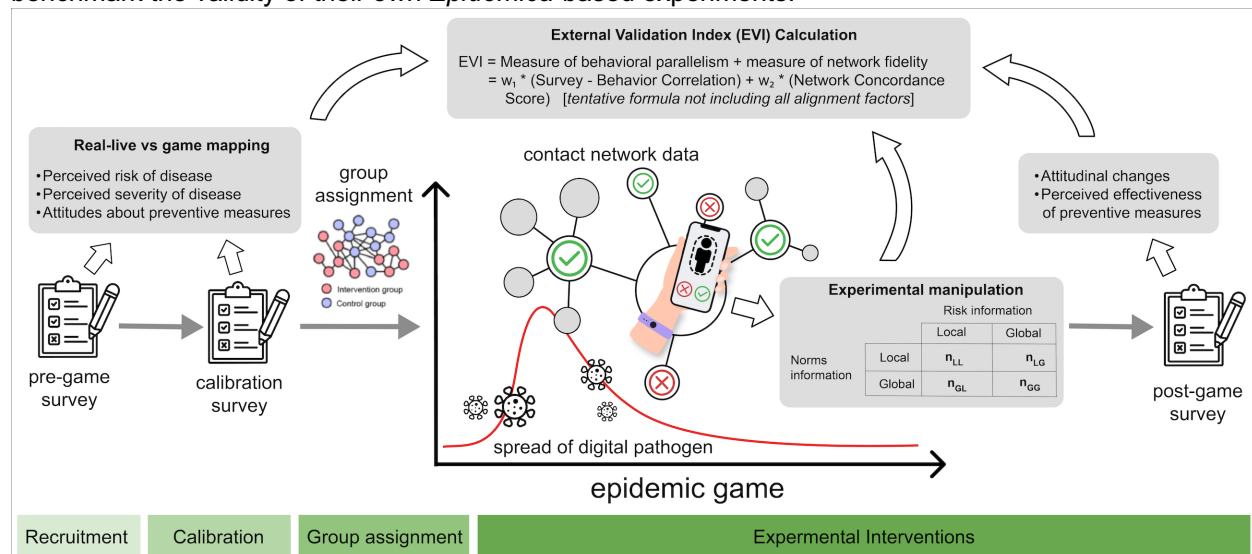


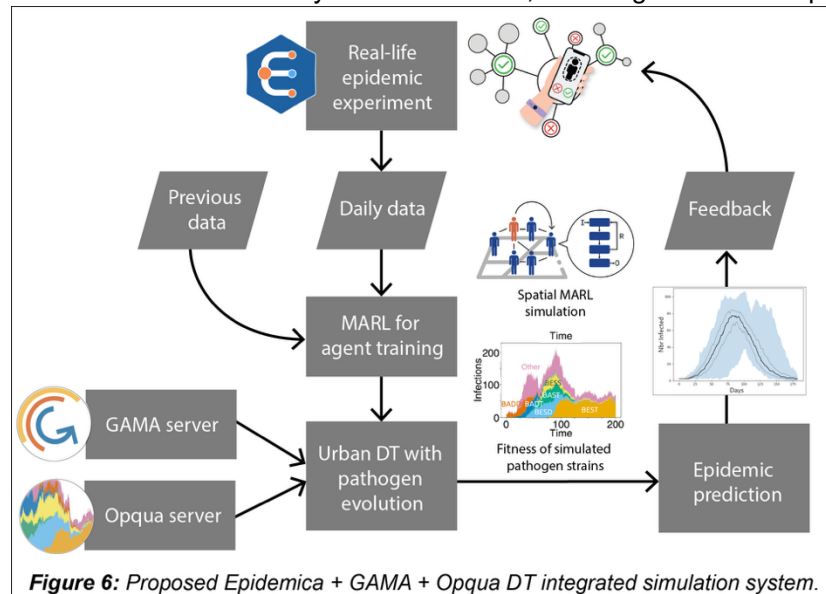
Figure 5: Diagram of epidemic game protocol, including its stages and overview of tentative external validation analysis (top box).

Diverse Participant Recruitment. While past epidemic games have focused on student populations in schools and colleges, we are now implementing targeted strategies to ensure data collection from diverse groups in the U.S. population. We are actively collaborating with community-engaged researchers such as Dr. Moley and Dr. Bowman, who have established connections with cultural centers, community organizations, and local public health networks. We will also leverage platforms such as ResearchMatch [104] to recruit a broader national sample, enabling participation across geographic, cultural, and socioeconomic backgrounds. These efforts will allow us to examine the external validity of behavioral responses across demographic and institutional contexts, improving generalizability of the validation framework and expanding the reach of *Epidemica*-based research.

Outcomes. This aim will produce a validated, general-purpose framework for behavioral and network data evaluation within *Epidemica*, applicable to future studies on compliance with public health interventions, peer influence, contact tracing, and more. It will also yield a replicable multi-site study protocol and the first systematic benchmarks of behavioral fidelity in a gamified epidemic simulation.

AIM 3: Integrate Urban Digital Twins with Evolving Synthetic Pathogens in Naturalistic Epidemic Experiments

Understanding the co-evolution of pathogen traits and human behavior under real-world constraints is essential to preparing for future epidemics [105-111]. Yet, existing models of infectious disease dynamics do not incorporate feedback loops between individual-level decisions, evolving pathogen traits, and city-scale interventions. Urban Digital Twins (DTs)—virtual replicas of cities integrating real-time spatial, behavioral, and infrastructural data—offer a powerful but underutilized opportunity to bridge this gap [112, 113]. In this aim, we propose to couple *Epidemica*-based experimental epidemic games with a city-scale DT and a synthetic pathogen evolution engine. This system will showcase the potential of such novel integration of experimental and computational methodologies and will allow us to simulate complex epidemic scenarios in real-time, enabling researchers to test policies and study feedback between individual decisions and systemic outcomes, including the effect on pathogen fitness.



Methodology. We will build a three-tiered simulation loop that connects real-world experimental data collected through *Epidemica*, a DT environment created using GAMA [114], an open-source modeling and simulation environment for creating spatially explicit agent-based simulations, and an pathogen evolution engine based on Opqua [115], an open-source epidemiological modeling framework that stochastically simulates pathogen transmission and evolution—modeling distinct genotypes spreading through host populations with mutation, recombination, and selection

dynamic (Figure 6). Our goal is to iteratively synchronize empirical data and simulations at multiple spatial and temporal scales. We have received LOCs from experts across these techniques and tools to ensure that my lab has appropriate guidance to develop and deploy this system successfully.

1. Construct a bi-directional *Epidemica*–Digital Twin pipeline. This pipeline begins with a data ingestion step, where participant mobility, interaction, and behavioral data are collected during *Epidemica* games (e.g., testing decisions, quarantine compliance) to seed the DT model. This data will then allow us to construct an empirically informed behavioral abstraction, where real-world behaviors will be transformed into agent archetypes in the GAMA environment via multi-agent reinforcement learning (MARL), enabling simulation of a broader synthetic population (up to millions of agents) beyond the participant group (in the order of hundreds or thousands). We will leverage ongoing development efforts that will make it possible to integrate general multi-agent reinforcement learning algorithms into GAMA through the PettingZoo library [116], and explore both offline [117] and online [53] MARL approaches to use real-time (daily) data and/or historical data accumulated over the duration of the epidemic game. The DT will layer environmental conditions (e.g., land use, transit flow, weather, infrastructure) over real contact network data to extrapolate large-scale spread dynamics.

2. Integrate synthetic pathogen evolution using Opqua. The *Epidemica*-based contact networks will serve as the substrate for daily infection and mutation events modeled by Opqua. We will use a modified

version of Opqua, where we will simulate pathogen transmission, strain dynamics, and selection pressures at the intra- and inter-host levels at sufficient throughput to feed the larger DT simulation and feedback loop. Genomic features of circulating strains will be updated across simulation cycles and mirrored within the *Epidemica* app to drive adaptation in participant behavior (e.g., higher perceived risk, symptom severity feedback) in response to changes in pathogens' virulence/infectivity

3. Establish a real-time feedback loop between real and synthetic epidemics. The outputs from the DT (e.g., synthetic epidemic size, resource saturation, emergence of high-transmissibility strains) will dynamically update game parameters in *Epidemica* for the next experimental round. These updates will include tailored intervention strategies (e.g., targeted messaging, simulated treatment availability) and constraints on movement or behavior, based on DT-predicted public health stress.

4. Test the system in pilot sites and evaluate cross-scale concordance. We will setup pilot deployments will take place in campus or city-scale settings where high-resolution spatial data are available, leveraging the collaborations we have already established in earlier stages of the project. We will analyze concordance/consistency between predicted DT dynamics and empirical *Epidemica* observations over time, which will inform updates in either the epidemic game or the DT model. Also, agent behavior and strain evolution will be assessed for emergent patterns, tipping points, and other features across scales.

Outcomes. This work will result in a truly novel integrative framework for modeling epidemic dynamics that links empirical human behavior, city-scale systems, and pathogen evolution. New insights into how behavioral adaptation shapes the evolutionary trajectory of pathogens and vice versa. Demonstration of Digital Twin-informed interventions that adapt in real time to behavioral and epidemiological dynamics. We will release open-source modules and APIs connecting *Epidemica* with GAMA and Opqua, enabling replication and reuse by the broader community.

8. COMMUNICATION AND DISSEMINATION

The *Epidemica* platform will be disseminated broader scientific community through a multi-pronged approach designed to maximize accessibility, visibility, and impact. We will publish results in high-impact, peer-reviewed journals in fields such as epidemiology, network science, behavioral sciences, and digital health. Manuscripts will emphasize the novel capabilities of *Epidemica* and report on the findings from use cases. A first manuscript introducing the platform is currently under preparation [118]. My lab, in partnership with collaborators, will organize workshops and webinars to demonstrate *Epidemica*'s capabilities to researchers, public health officials, and educators, including at a high-profile event in the field such as the Pandemic Sciences Conference, which will be hosted outside of Oxford from next 2026 onwards. Hands-on training sessions will cover study design, data collection, and data analysis using the platform. We will setup a source-code repository where regulars release and documentation will be made available. *Epidemica*'s software components, including data collection modules, template app, and backed infrastructure, will be released as open-source software under the MIT License to ensure broad accessibility. Comprehensive documentation will be provided, including a user manual, API documentation, and sample scripts for integration into R and Python analysis pipelines. All these materials will be accessible through the project's main page [119], where an online community portal will be established, including discussion forums, FAQs, and knowledge base articles. The portal will facilitate peer support, sharing of best practices, and collaborative study design.

9. OUTCOMES ASSESSMENT

For intellectual merit assessment, we will consider platform development and functionality (including successful deployment of *Epidemica*-based apps, meeting predefined technical specifications and user requirements, completion of platform testing and validation across diverse study designs), scientific contributions (number of peer-reviewed publications, conference presentations, and citations arising from research conducted using *Epidemica*, quality of research outputs, as evaluated by expert peer reviewers and collaborators as well as by own data quality and FAIRness metrics), and research impact by adoption of *Epidemica* by NSF-funded and other research groups for studies on behavior, networks, and disease transmission. Evidence of *Epidemica* advancing research in digital epidemiology, behavioral science, and network theory, gathered from the published literature and exchanges with our core collaborators and other experts. With regards to broader impacts assessment, we will evaluate our outcomes according to the translational impact of *Epidemica* studies, i.e.: whether they lead to policy-relevant applications and

insights, and open science and accessibility, e.g.: number of downloads and instances of *Epidemica* software, active users and contributions to the online community portal, and pen-source contributions and use of the platform in diverse research settings.

10. PROJECT TIMELINE

Aim 1 should be carried out continuously for the first 4 years, since platform development is expected to happen all along (but with efforts from dev team front loaded in the first year to build out the basic modules of the *Epidemica* infrastructure). Aim 2, which centers on the validation of behavioral and network data through experimental games, will be carried out in Years 2 and 3, once the core platform is operational. Aim 3, integrating *Epidemica* with Digital Twin models and evolving synthetic pathogens, will be implemented in Years 2 through 4. Year 5 will be dedicated to dissemination activities, including the publication of results, deployment of reusable tools and documentation, and community engagement to broaden the impact and adoption of the platform.

	2026	2027	2028	2029	2030
Aim 1: Platform Development (core modules)					
Aim 1: Platform Development (continuous support))					
Aim 2: Behavioral & Network Validation Studies					
Aim 3: Digital Twin & Pathogen Evolution Experiments					
Dissemination of results					

11. BROADER IMPACTS

We envisage our work with *Epidemica* having several broader impacts beyond addressing the lack of infrastructure to generate data for experimental epidemiology, as the data from *Epidemica* informs better models and enable the development of new tools by the community.

Policy impact and pandemic preparedness. *Epidemica*'s infrastructure will provide unique information on the structure of real-life contact networks, a deeper understanding of the relative influence of environmental factors on pathogen transmission dynamics and the potential effectiveness of public health policies under different conditions. Our platform will also significantly advance the potential for infectious disease experimentation, which has been difficult and resource-consuming to run in real-life contexts [120], and it will open a new avenue for research on NPIs as well as studies for epidemic preparedness.

Educational outreach from the local to global stage. Given my role as co-founder of the OO project with collaborators from the Broad Institute of Harvard and MIT [2, 121], I anticipate a positive synergy between the OO and *Epidemica* project. Some technology elements of OO could benefit from modules that will be refined further as part of *Epidemica*'s aims and, therefore, help improve the performance and robustness of the OO app. This would have an impact on the educational outreach that is the main mission of the OO project. Some of *Epidemica* experiments could also have a secondary educational impact, as some of those experiments could be inspired on the model we developed with the use of the OO app in college campuses, where students not only participate in the experiments as subjects but also contribute to their planning and implementation, as well as to the data analysis. During and after the experiments, students will engage as participants, organizers, and researchers. We anticipate writing and publishing joint manuscripts with those institutions who are interested in the research and knowledge dissemination.

Empirically informed Urban Digital Twin + Reinforcement Learning Framework. While our proposed DT framework has a focus on studying epidemic process and pathogen adaptation to behavioral change, this is, in our opinion, a highly innovative approach for other kinds of social and behavioral studies that would benefit of high-resolution DT models where agents are trained with real-time empirical data through RL.

Launch a global consortium for developing and deploying digital tools for pandemic response. Given the track record of my collaborators in the U.K. on digital contact tracing and epidemic games using mobile apps, as well as the broad expertise of the core collaborators across network theory, social sciences, and more, we anticipate this proposal to be the start of a transdisciplinary technology and scientific foundation and global consortium for improve pre-pandemic planning and pandemic response using the tools that we will be prototyping, testing, and evaluating through various experimental designs in *Epidemica*.

12. RESULTS FROM PRIOR NSF FUNDING

PI Colubri has no prior NSF funding.

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